

Rabi Oscillations

In these notes we will apply the Dyson series to a simple example of a two-level system, which is the simplest non-trivial quantum system. We will see that the Dyson series allows us to compute the transition amplitude for a particle to go from one state to another in the presence of a time-dependent perturbation, and that this transition amplitude exhibits oscillations known as Rabi oscillations. In Sakurai's book, this problem is solved exactly for a circularly polarized field. Here we will solve it perturbatively using the Dyson series, and then show how the exact solution can be obtained by “resumming” the Dyson series.

The full Hamiltonian of the system is given by

$$\hat{H}(t) = \frac{\omega_0}{2} \sigma_z + \frac{\Omega}{2} \cos(\omega t) \sigma_x + \frac{\Omega}{2} \sin(\omega t) \sigma_y, \quad (1)$$

where we will assume that $\Omega \ll \omega_0$ so that we can treat the second and third terms as a perturbation. In our notation, we have

$$\hat{H}_0 = \frac{\omega_0}{2} \sigma_z = \begin{pmatrix} \frac{\omega_0}{2} & 0 \\ 0 & -\frac{\omega_0}{2} \end{pmatrix}, \quad (2)$$

and a perturbation that can be written in a couple of ways

$$\hat{V}(t) = \frac{\Omega}{2} \cos(\omega t) \sigma_x + \frac{\Omega}{2} \sin(\omega t) \sigma_y = \frac{\Omega}{2} \sigma_+ e^{-i\omega t} + \frac{\Omega}{2} \sigma_- e^{i\omega t} = \frac{\Omega}{2} \begin{pmatrix} 0 & e^{-i\omega t} \\ e^{i\omega t} & 0 \end{pmatrix}. \quad (3)$$

We used $\sigma_{\pm} = \frac{1}{2}(\sigma_x \pm i\sigma_y)$ in the second equality. The energy of the unperturbed system is given by $E_{\pm} = \pm \frac{\omega_0}{2}$, and the corresponding eigenstates are

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (4)$$

respectively. Let us define the interaction picture states and operators as follows:

$$|\psi_I(t)\rangle = e^{\frac{i}{\hbar} \hat{H}_0 t} |\psi(t)\rangle = \begin{pmatrix} e^{\frac{i}{2} \omega_0 t} & 0 \\ 0 & e^{-\frac{i}{2} \omega_0 t} \end{pmatrix} |\psi(t)\rangle, \quad (5)$$

and

$$\hat{V}_I(t) = e^{\frac{i}{\hbar} \hat{H}_0 t} \hat{V}(t) e^{-\frac{i}{\hbar} \hat{H}_0 t} = \frac{\Omega}{2} \begin{pmatrix} 0 & e^{-i\delta t} \\ e^{i\delta t} & 0 \end{pmatrix}, \quad (6)$$

where $\delta = \omega - \omega_0$ is called the detuning.

Transitions at the resonance We would like to compute the transition amplitude for the system to go from the state $|+\rangle$ at time $t = 0$ to the state $|-\rangle$ at time t . First, it must be clear that if $\omega = 0$, then the system will remain in whatever state it starts in, and so the transition amplitude will be zero. In other words, $\hat{H}_0$ is diagonal in the $|\pm\rangle$ basis, and so it does not mix the two states. When the condition $\delta = 0$ ($\omega = \omega_0$) is satisfied, we say that the system is on resonance, and we will see that this is when the transition amplitude is maximized.

At the resonance,

$$\hat{V}_I(t) = \frac{\Omega}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\Omega}{2} \sigma_x, \quad (7)$$

is time-independent! In this regime, the interaction picture time-evolution of the states is easily exponentiated,

$$|\psi_I(t)\rangle = e^{-\frac{i}{\hbar} \hat{V}_I t} |\psi_I(0)\rangle = e^{-\frac{i}{\hbar} \frac{\Omega}{2} \sigma_x t} |\psi_I(0)\rangle = \left[\cos\left(\frac{\Omega t}{2}\right) - i \sin\left(\frac{\Omega t}{2}\right) \sigma_x \right] |\psi_I(0)\rangle, \quad (8)$$

where we used the matrix exponential identity $e^{i\alpha\hat{n}\cdot\vec{\sigma}} = \cos(\alpha) + i\sin(\alpha)\hat{n}\cdot\vec{\sigma}$ for any unit vector $\hat{n}$. From this expression, we can immediately read off the transition amplitude for the system to go from $|+\rangle$ to $|-\rangle$ at time t ,

$$\langle - | \hat{U}_I(t) | + \rangle = \langle - | \psi_I(t) \rangle = -i \sin\left(\frac{\Omega t}{2}\right), \quad (9)$$

or in terms of the probability,

$$\boxed{P_{+\rightarrow-}(t) = \sin^2\left(\frac{\Omega t}{2}\right)}, \quad (10)$$

which exhibits maximal oscillations with frequency $\Omega/2$ (the amplitude is 1 right at $t = \pi/\Omega$).

Transitions with detuning (exact solution)

When $\delta \neq 0$, the interaction picture potential in Eq. (6) is time-dependent, and so we cannot simply exponentiate it to find the time-evolution of the states. However, there is a trick we can use to find the exact solution. We go to a rotating frame (the tilde frame) that eliminates the remaining time-dependence in $\hat{V}_I(t)$. (Note: this is *not* the rotating wave approximation, which refers to dropping counter-rotating terms from a linearly polarized driving field. Here, the Hamiltonian already contains only a single circularly polarized component, and the frame transformation we perform is exact.)

We apply the unitary transformation

$$\hat{U}_R(t) = e^{-\frac{i}{2}\delta t \sigma_z} = \begin{pmatrix} e^{-\frac{i}{2}\delta t} & 0 \\ 0 & e^{\frac{i}{2}\delta t} \end{pmatrix}, \quad \text{so that} \quad |\psi_I(t)\rangle = \hat{U}_R(t) |\tilde{\psi}_I(t)\rangle. \quad (11)$$

In this rotating frame, the potential becomes time-independent,

$$\tilde{V} = \hat{U}_R^\dagger(t) \hat{V}_I(t) \hat{U}_R(t) = \frac{\Omega}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\Omega}{2} \sigma_x. \quad (12)$$

Since the interaction picture Schrödinger equation is $i\hbar \partial_t |\psi_I\rangle = \hat{V}_I |\psi_I\rangle$ (only the interaction Hamiltonian appears), the equation of motion for $|\tilde{\psi}_I\rangle$ picks up an extra term from the time-dependent frame:

$$i\hbar \partial_t |\tilde{\psi}_I\rangle = \tilde{H} |\tilde{\psi}_I\rangle, \quad \tilde{H} = \hat{U}_R^\dagger \hat{V}_I \hat{U}_R - i\hbar \hat{U}_R^\dagger \dot{\hat{U}}_R = -\frac{\delta}{2} \sigma_z + \frac{\Omega}{2} \sigma_x. \quad (13)$$

This Hamiltonian is time-independent, and so we can easily exponentiate it to find the time-evolution of the states in the rotating frame, and then transform back to the original frame to find the transition amplitude.

First, note that the eigenvalues of $\tilde{H}$ are $\pm \frac{\Omega_R}{2}$, with Ω_R called the generalized Rabi frequency:

$$\Omega_R \equiv \sqrt{\delta^2 + \Omega^2}. \quad (14)$$

The corresponding eigenstates are given by

$$|+\rangle_R \equiv \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad |-\rangle_R \equiv \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}, \quad (15)$$

where $\tan(2\theta) = -\Omega/\delta$. The time-evolution of the states in the rotating frame is then given by

$$|\tilde{\psi}_I(t)\rangle = e^{-\frac{i}{\hbar} \tilde{H} t} |\tilde{\psi}_I(0)\rangle = \left[\cos\left(\frac{\Omega_R t}{2}\right) - i \sin\left(\frac{\Omega_R t}{2}\right) \left(-\frac{\delta}{\Omega_R} \sigma_z + \frac{\Omega}{\Omega_R} \sigma_x\right) \right] |\tilde{\psi}_I(0)\rangle. \quad (16)$$

Transforming back to the interaction picture, $|\psi_I(t)\rangle = \hat{U}_R(t) |\tilde{\psi}_I(t)\rangle$, we find that the transition amplitude for the system to go from $|+\rangle$ to $|-\rangle$ at time t is given by

$$\langle - | \hat{U}_I(t) | + \rangle = \langle - | \hat{U}_R(t) | \tilde{\psi}_I(t) \rangle = -ie^{i\delta t/2} \frac{\Omega}{\Omega_R} \sin\left(\frac{\Omega_R t}{2}\right). \quad (17)$$

Finally, the transition probability is given by the exact formula

$$P_{+\rightarrow-}(t) = \frac{\Omega^2}{\delta^2 + \Omega^2} \sin^2\left(\frac{\sqrt{\delta^2 + \Omega^2} t}{2}\right). \quad (18)$$

which recovers Sakurai's result for the transition probability in the presence of a time-dependent perturbation, and reduces to the previous result at the resonance ($\delta = 0$). Note, however, that if we are off the resonance by a lot ($\delta \gg \Omega$), then the transition probability is suppressed by a factor of Ω^2/δ^2 , and so the system will remain in the state it started in with high probability.



The rotating wave approximation is a separate technique, used when the driving field is *linearly* polarized. In that case, one writes $\cos(\omega t)\sigma_x = \frac{1}{2}(e^{-i\omega t} + e^{+i\omega t})\sigma_x$ and drops the counter-rotating term oscillating at frequency $\omega + \omega_0$. This approximation is valid when $\Omega \ll \omega + \omega_0$, which is typically the case in atomic physics and quantum optics, where the driving frequency ω is close to the transition frequency ω_0 of the system. We will return to this in the solved problems or homeworks.

Transitions with detuning (perturbative)

Let us apply our perturbation theory formalism to this problem. First, note that the zeroth-order transition amplitude is given by

$$A_{+\rightarrow-}^{(0)}(t) = \langle - | \mathbb{1} | + \rangle = 0, \quad (19)$$

No transition occurs without at least a single insertion of the potential. Now, the first-order transition amplitude is more interesting. Before proceeding, let us define the phase

$$\phi = \frac{t\delta}{2} \quad (20)$$

and the “amplitude”

$$\theta = \frac{\Omega}{\delta}. \quad (21)$$

Then, the first-order transition amplitude is given by

$$A_{+\rightarrow-}^{(1)}(t) = -i \int_0^t dt' \langle - | \hat{V}_I(t') | + \rangle = -i \int_0^t dt' \frac{\Omega}{2} e^{i\delta t'} = -\frac{\Omega}{2} \frac{e^{i\delta t} - 1}{\delta} \quad (22)$$

$$= -i\theta \sin(\phi) e^{i\phi} \simeq -\frac{i\Omega t}{2} = -\frac{i\theta\phi}{2} \text{ for } \phi \ll 1. \quad (23)$$

Note that at the resonance ($\delta = 0$ with $\phi = 0$), the first-order transition is just the small angle approximation of the exact result in Eq. (9). The amplitude grows linearly with time, and so the transition probability grows quadratically with time, as expected from time-dependent perturbation theory.

Indeed, the transition probability at first order, including δ , is

$$P_{+\rightarrow-}^{(1)}(t) = \left| A_{+\rightarrow-}^{(0)}(t) + A_{+\rightarrow-}^{(1)}(t) \right|^2 = \theta^2 \sin^2(\phi). \quad (24)$$

This is an oscillation with frequency $\delta/2$, and amplitude θ^2 . The perturbative solution is only valid for short times, nevertheless, it reveals something curious: at short times, the characteristic oscillation frequency of the system is given by the detuning δ , and not by the Rabi frequency Ω .

Let us also check the second order result. It turns out the second-order transition amplitude vanishes exactly:

$$A_{+\rightarrow-}^{(2)}(t) = (-i)^2 \int_0^t dt' \int_0^{t'} dt'' \langle - | \hat{V}_I(t') \hat{V}_I(t'') | + \rangle = 0. \quad (25)$$

This is a consequence of the fact that V_I^2 is a diagonal matrix, and so it cannot mix the two states. You can easily check this in the $\delta = 0$ case, where $V_I = \frac{\Omega}{2} \sigma_x$, and so $V_I^2 = \frac{\Omega^2}{4} \mathbb{1}$. Clearly, any even order in our perturbation theory will vanish by the same argument! This is not surprising; the exact solution only has odd powers of Ω in the expansion of the sine function, and so we should expect that all even orders vanish.

For $n = 3$, we have

$$V_I(t_3)V_I(t_2)V_I(t_1) = \left(\frac{\Omega}{2}\right)^3 \begin{pmatrix} 0 & e^{-i\delta(t_3-t_2+t_1)} \\ e^{i\delta(t_3-t_2+t_1)} & 0 \end{pmatrix} \quad (26)$$

and so the third-order transition amplitude is given by (sparing you the details)

$$A_{+\rightarrow-}^{(3)}(t) = (-i)^3 \left(\frac{\Omega}{2}\right)^3 \int_0^t dt_3 \int_0^{t_3} dt_2 \int_0^{t_2} dt_1 e^{i\delta(t_3-t_2+t_1)} = i \frac{\Omega^3}{2\delta^3} \left[\sin \frac{t\delta}{2} - \left(\frac{t\delta}{2}\right) \cos \frac{t\delta}{2} \right] e^{i\delta t/2} \quad (27)$$

$$= i \frac{\theta^3}{2} (\sin(\phi) - \phi \cos(\phi)) e^{i\phi} \simeq \frac{i\theta^3\phi^3}{6} \text{ for } \phi \ll 1. \quad (28)$$

Adding this to the zeroth and first-order result, we find the transition amplitude to third order. It is not very insightful at this stage, so I proceed.

For $n = 5$, **Mathematica** has us covered and we can quote the result for the transition amplitude to fifth order:

$$A_{+\rightarrow-}^{(5)}(t) = -\frac{i}{8} \theta^5 [(3 - \phi^2) \sin \phi - 3\phi \cos \phi] e^{i\phi} \simeq -\frac{i\theta^5\phi^5}{5!} \text{ for } \phi \ll 1. \quad (29)$$

Okay, let us put all this together. To fifth order, the transition amplitude is given by

$$A_{+\rightarrow-}(t) = -ie^{i\phi} \left[\theta \sin \phi - \frac{\theta^3}{2} (\sin \phi - \phi \cos \phi) + \frac{\theta^5}{8} [(3 - \phi^2) \sin \phi - 3\phi \cos \phi] + \dots \right] \quad (30)$$

which in this form is quite a messy result. Let us investigate the early time regime, where it simplifies considerably:

$$A_{+\rightarrow-}(t) \simeq -i \left(\theta\phi - \frac{\theta^3\phi^3}{3!} + \frac{\theta^5\phi^5}{5!} + \dots \right) e^{i\phi} = -ie^{i\phi} \sin(\theta\phi) = -ie^{i\delta t/2} \sin\left(\frac{\Omega t}{2}\right), \quad (31)$$

This is exactly the early time expansion of the exact result at the resonance Eq. (9), which is quite remarkable. This procedure of summing up the perturbative series to find the exact result is called *resummation*, and it is a very powerful technique when it works; What we are doing here is including all terms of our Dyson series and showing that it indeed converges to the exact result. The challenge is of course that not all perturbative series can be as easily resummed. Higher order terms may not add up to a simple functional form as we have here, but at least we have verified our method with a simple example.

For the general case of late-time oscillations, I would like to go back to Eq. (18) and try to rewrite it in a form that makes the perturbative expansion more manifest. We write the exact amplitude (and bar it to denote the dropping of the overall phase $-ie^{i\phi}$) as (drumrolls)

$$\bar{A}_{+\rightarrow-}^{\text{exact}}(t) = \frac{\theta}{\sqrt{1+\theta^2}} \sin\left(\sqrt{1+\theta^2} \phi\right) = \left(\theta - \frac{\theta^3}{2} + \dots\right) \left(\sin \phi + \frac{\theta^2 \phi}{2} \cos \phi + \dots\right), \quad (32)$$

$$= \theta \sin \phi - \frac{\theta^3}{2} (\sin \phi - \phi \cos \phi) + \dots, \quad (33)$$

$$\simeq \theta \phi - \frac{\theta^3 \phi^3}{3!} + \dots \text{ for } \phi \ll 1, \quad (34)$$

Again, the same expansion structure as in our perturbative result arises. We would be pressed to find this expansion structure without the exact result.